

# Probability Distribution

# Random Variable

- A variable used to denote the numerical value of the outcome of an experiment is called the random variable, abbreviated as r.v.
- A random variable is denoted by capital letters  $X, Y, Z, \dots$  and its values are denoted by  $x_1, x_2, \dots, y_1, y_2, \dots, z_1, z_2, \dots$  etc.

# Discrete Random Variable

- A random variable is called a **discrete random variable** if it takes discrete (distinct) values

$x_1, x_2, \dots x_n \dots in (a, b).$

For example, number of floors in a building, number of kids in a family etc.

# Continuous Random Variable

- A random variable is called a **continuous random variable** if it takes all values between an interval  $(a, b)$ .
- For example, age, height, weight are continuous random variables.

# Probability Distribution Of A discrete Random Variable

- Let  $X$  be a **discrete random variable** .
- Let  $x_1, x_2, \dots, x_n, \dots$  be the possible values of  $X$ .  
with each possible outcome  $x_i$  we associate a number  $p(x_i) = p(X = x_i) = p_i$  called the Probability of  $x_i$ .

- The numbers  $p(x_i), i = 1, 2, \dots, n \dots$  must satisfy the following conditions :
  1.  $p(x_i) \geq 0$  for all  $i$
  2.  $\sum_{i=1} p(x_i) = 1$ .
- The function  $p$  is called the **probability function** or **probability mass function (p.m.f.)** or **probability density function (p.d.f.)** of the random variable  $X$  and the set of pairs  $(x_i, p_i)$  is called the probability distribution of  $X$ .

- The probability distribution of a discrete random variable  $X$  taking values  $x_1, x_2, x_3 \dots x_n \dots$  with probabilities  $p_1, p_2, p_3 \dots p_n \dots$  where  $p_1 \geq 0$  and  $\sum p_i = 1$  can be given in tabular form as.

|              |       |       |                         |                         |
|--------------|-------|-------|-------------------------|-------------------------|
| $X$          | $x_1$ | $x_2$ | $x_3 \dots \dots \dots$ | $x_n \dots \dots \dots$ |
| $P(X = x_i)$ | $p_1$ | $p_2$ | $p_3 \dots \dots \dots$ | $p_n \dots \dots \dots$ |

# Distribution Function of A Discrete Random Variable X

- Suppose, X is a random variable taking values  $x_1, x_2 \dots x_n$  with probabilities

$p(x_i), i = 1, 2, \dots n \dots$  such that

(i)  $p(x_i) \geq 0$  for all  $i$ ,      (ii)  $\sum p(x_i) = 1$

Consider F defined by

$$F(x_i) = P(X \leq x_i), i = 1, 2, 3, \dots$$

i.e.  $F(x_i) = P(x_1) + P(x_2) + \dots + P(x_i)$

then the function F is called the

**cumulative distribution function**

**or simply distribution function**



- The set of pairs  $\{x_i, F(x_i)\}$  is called the **cumulative probability distribution**.
- Consider the following table

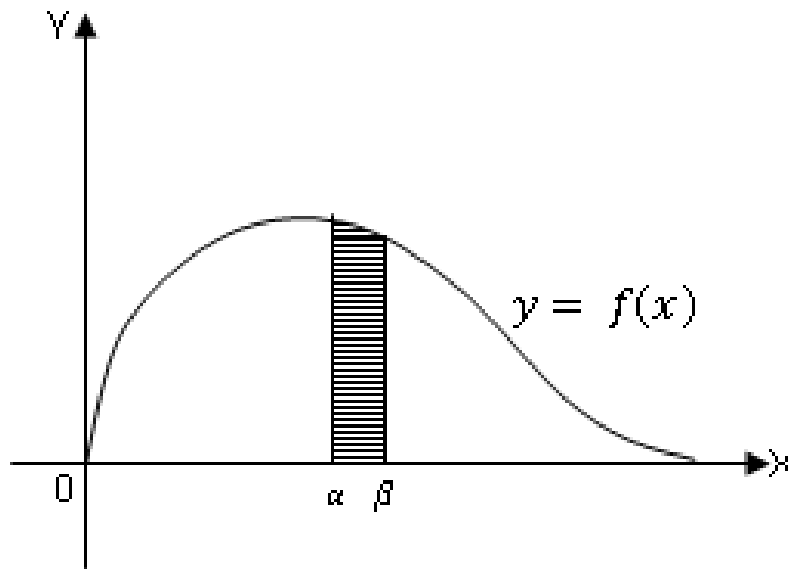
| $X$                   | $x_1$    | $x_2$             | $x_3 \dots\dots\dots$        | $x_n \dots\dots\dots$        |
|-----------------------|----------|-------------------|------------------------------|------------------------------|
| $F(x_i) = F(X = x_i)$ | $P(x_1)$ | $\sum_1^2 P(x_i)$ | $\sum_1^3 P(x_i) \dots\dots$ | $\sum_1^n P(x_i) \dots\dots$ |

The function  $F$  is called the **distribution function**

# Probability Density Function Of A Continuous Random Variable

- A continuous function  $y = f(x)$  such that
    - (i)  $f(x)$  is integrable.
    - (ii)  $f(x) \geq 0$
    - (iii)  $\int_a^b f(x)dx = 1$  if  $X$  lies in  $[a, b]$  and
    - (iv)  $\int_\alpha^\beta f(x)dx = P(\alpha \leq X \leq \beta)$   
where  $a < \alpha < \beta < b$
- is called **probability density function** of a continuous random variable  $X$ .

- The curve given by  $y = f(x)$  is called the **probability density curve** or simply **probability curve**.
- The expression  $f(x)dx$  is usually denoted by  $df(x)$  and is known as **probability differential**



- You know that for discrete random variable the probability at  $X = C$  may not be zero.
- But, in a continuous random variable  $P(X = C)$  is always zero because

$$P(X = C) = \int_C^C f(x)dx$$

and this definite integral is zero.

- For a continuous random variable  $X$ 

$$P(\alpha \leq X \leq \beta) = P(\alpha < X < \beta)$$

$$= P(\alpha < X \leq \beta) = P(\alpha \leq X < \beta)$$

In other words we may include or may not include the end points in the interval.

# Continuous Distribution Function

- If  $X$  is a continuous random variable  $X$ , having the probability density function  $f(x)$  then the function

$$\begin{aligned} F(x) &= P(X \leq x) \\ &= \int_{-\infty}^x f(t) dt, -\infty < x < \infty \end{aligned}$$

is called **distribution function** or **cumulative distribution function** of the random variable  $X$ .

# Expectation of a Random Variable

- If a discrete random variable  $X$  assumes values  $x_1, x_2 \dots x_n \dots$  with probability  $P_1, P_2 \dots P_n \dots$  respectively then the mathematical expectation of  $X$  denoted by  $E(x)$  is defined by

$$\begin{aligned} E(X) &= P_1X_1 + P_2X_2 + \dots + P_nX_n \dots \\ &= \sum P_iX_i \quad \text{where } \sum P_i = 1. \end{aligned}$$

This value is also referred to as **mean** value of  $X$ .

- Let  $X$  be a continuous random variable with probability density function  $f(x)$ .

Then the mathematical expectation of  $X$ , denoted by  $E(X)$  is defined by

$$E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

$$\text{where, } \int_{-\infty}^{\infty} f(x) dx = 1$$

# Laws of Expectation

- If  $X$  is a discrete random variable such that  $x_i \geq 0$  for all  $i$ , then  $E(X) \geq 0$
- If  $X$  is a discrete (or continuous) random variable,  $a$  and  $b$  are constants then  $E(aX + b) = aE(X) + b$
- Putting  $a = 0$ .  $E(b) = b$   
i.e. expectation of a constant is the constant itself.
- Putting  $b = 0$ .  $E(ax) = aE(X)$   
i.e. for calculations the constant can be taken out.



- Putting  $a = 1, b = -\bar{X}$ ,  $E(X - \bar{X}) = 0$ .

- **Theorem of Addition:**

The expectation of the sum (or difference) of two (discrete or continuous) variates is equal to the sum (or difference) of their expectations.

In symbols,  $E(X \pm Y) = E(X) \pm E(Y)$

- **Theorem of Multiplication:**

The expectation of the product of **two independent** variates (discrete or continuous) is equal to the product of their expectations if the expectation exist.

- In symbols,  $E(XY) = E(X).E(Y)$

**Note:** It should be noted that the converse of the above theorem is not. If  $E(XY) = E(X).E(Y)$  then it does not mean that X, Y are independent.

# VARIANCE

- $\text{Var}(X) = E(X - \bar{X})^2$   
 $= E[X - E(X)]^2$   
 $= E[X^2 - 2XE(X) + \{E(X)\}^2]$   
 $= E(X^2) - 2E(X) \cdot E(X) + [E(X)]^2$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

# Properties of Variance

- Variance of a constant is zero,  $V(C) = 0$ .
- If  $X$  is a random variate and  $a, b$  are constants  
then  $V(aX + b) = a^2V(X)$
- $V(aX) = a^2V(X)$
- $V(X + b) = V(X)$

- Note that  $E(aX + b) = aE(X) + b$   
we do not have  $V(aX + b) = aV(X) + b$ .  
Instead, we have  $V(aX + b) = a^2V(X)$ .
- $V(a_1X_1 + a_2X_2) = a_1^2V(X_1) + a_2^2V(X_2)$   
Where  $X_1$  and  $X_2$  are independent random variates.
- If  $a_1 = 1, a_2 = 1$ , we get  
 $V(X_1 + X_2) = V(X_1) + V(X_2)$  &
- If  $a_1 = 1, a_2 = -1$ , we get  
 $V(X_1 - X_2) = V(X_1) + V(X_2)$

EX.1 If  $X_1$  has mean 4 and variance 9 and  $X_2$  has mean  $-2$  variance 4, and the two are independent, find  $E(2X_1 + X_2 - 3)$  and  $V(2X_1 + X_2 - 3)$

**Solution:** We have  $E(X_1) = 4, V(X_1) = 9,$   
 $E(X_2) = -2$  and  $V(X_2) = 4$

$$\begin{aligned}\therefore E(2X_1 + X_2 - 3) &= E(2X_1 + X_2) - 3 \\ &= 2E(X_1) + E(X_2) - 3 \\ &= 2(4) + (-2) - 3 = 3\end{aligned}$$

$$\begin{aligned}\therefore V(2X_1 + X_2 - 3) &= V(2X_1 + X_2) \\ &= 2^2 V(X_1) + V(X_2) \\ &= 4(9) + 4 = 40\end{aligned}$$

# MEDIAN

- The median is the size of the item which lies at the middle
- For a continuous distribution the median  $M$  divides the area under the curve from  $x = a$  to  $x = b$  into two equal parts.
- If  $M$  is the median then  $\int_a^M f(x) dx = \int_M^b f(x) dx = \frac{1}{2}$
- By solving any one of the equations
$$\int_a^M f(x) dx = \frac{1}{2}, \quad \text{or} \quad \int_M^b f(x) dx = \frac{1}{2}$$
We can get the median  $M$ .

# MODE

- Mode is the size of the item having maximum frequency
- Using the theory of maxima, mode is obtained by solving the equation

$$\frac{dy}{dx} = 0 \text{ i.e. } f'(x) = 0 \text{ with the condition that}$$
$$\frac{d^2y}{dx^2} < 0 \text{ i.e. } f''(x) < 0 \text{ and that } x \text{ lies in the interval } [a, b] \text{ of } X.$$



EX.2 Given the following probability function of a discrete random variable  $X$

|            |   |     |      |      |      |       |        |            |
|------------|---|-----|------|------|------|-------|--------|------------|
| $x$        | 0 | 1   | 2    | 3    | 4    | 5     | 6      | 7          |
| $P(X = x)$ | 0 | $c$ | $2c$ | $2c$ | $3c$ | $c^2$ | $2c^2$ | $7c^2 + c$ |

- Find
- (i)  $c$
  - (ii)  $P(X \geq 6)$ ,
  - (iii)  $P(X < 6)$ ,
  - (iv)  $k$  if,  $P(X \leq k) > 1/2$ , where  $k \in N$ ,
  - (v)  $P(1.5 < X < 4.5/X > 2)$
  - (vi)  $E(X)$
  - (vii)  $V(X)$

**Solution:** Since  $\sum p_i = 1$

$$\therefore 0 + c + 2c + 2c + 3c + c^2 + 2c^2 + 7c^2 + c = 1$$

$$\therefore 10c^2 + 9c = 1$$

$$c = -1 \text{ or } 0.1$$

$c = -1$  is not possible as it represents probability

$$\therefore c = 0.1$$

Hence the probability distribution of X is

| X                | 0 | 1   | 2   | 3   | 4   | 5    | 6    | 7    |
|------------------|---|-----|-----|-----|-----|------|------|------|
| P<br>( $X = x$ ) | 0 | 0.1 | 0.2 | 0.2 | 0.3 | 0.01 | 0.02 | 0.17 |

$$\begin{aligned} \text{(ii)} \quad P(X \geq 6) &= P(X = 6) + P(X = 7) \\ &= 0.02 + 0.17 = 0.19 \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad P(X < 6) &= 1 - P(X \geq 6) \\ &= 1 - 0.19 = 0.81 \end{aligned}$$

(iv) To find  $k$  if,  $P(X \leq k) > 1/2$ , where  $k \in N$

$$P(X \leq 0) = 0$$

$$P(X \leq 1) = 0.1$$

$$P(X \leq 2) = 0.3$$

$$P(X \leq 3) = 0.5$$

$$P(X \leq 4) = 0.8$$

$$\therefore k = 4$$

$$\begin{aligned}
 \text{(v)} \quad & P(1.5 < X < 4.5 / X > 2) \\
 &= \frac{P[(1.5 < X < 4.5) \cap (X > 2)]}{P(X > 2)} \\
 &= \frac{P(2 < X < 4.5)}{P(X > 2)} \\
 &= \frac{P(X = 3) + P(X = 4)}{P(X > 2)} \\
 &= \frac{0.5}{0.7} = \frac{5}{7}
 \end{aligned}$$

$$\begin{aligned}
 \text{(vi) } E(X) &= \sum p_i x_i \\
 &= 0 \times 0 + 1 \times 0.1 + 2 \times 0.2 + 3 \times 0.2 + 4 \times 0.3 + 5 \\
 &\quad \times 0.01 + 6 \times 0.02 + 7 \times 0.17 \\
 &= 3.66
 \end{aligned}$$

(vii) To find  $V(X)$

$$\begin{aligned}
 E(X^2) &= \sum p_i x_i^2 \\
 &= 1 \times 0.1 + 4 \times 0.2 + 9 \times 0.2 + 16 \times 0.3 + 25 \times 0.01 \\
 &\quad + 36 \times 0.02 + 49 \times 0.17 \\
 &= 16.8
 \end{aligned}$$

$$\begin{aligned}
 V(x) &= E(X^2) - [E(X)]^2 \\
 &= 16.8 - (3.66)^2 \\
 &= 3.4044
 \end{aligned}$$

EX.3 The probability distribution of a discrete random variable is given by

|            |     |     |     |     |
|------------|-----|-----|-----|-----|
| $X$        | 0   | 1   | 2   | 3   |
| $P(X = x)$ | $k$ | 0.3 | 0.5 | $k$ |

If  $Y = X^2 + 2X$ , find the probability distribution of  $Y$ , mean and variance of  $Y$ .

What is the probability that  $1 < y < 10$ ?

**Solution:** Since  $\sum p_i = 1$

$$k + 0.3 + 0.5 + k = 1$$

$$k = 0.1$$

Hence, the p.d.f. of  $X$  is

| $X$        | 0   | 1   | 2   | 3   | Total |
|------------|-----|-----|-----|-----|-------|
| $P(X = x)$ | 0.1 | 0.3 | 0.5 | 0.1 | 1     |

Now, when  $X$  takes values 0, 1, 2, 3.

$Y = X^2 + 2X$  takes values 0, 3, 8, 15 resply  
with respective probabilities

|            |     |     |     |     |       |
|------------|-----|-----|-----|-----|-------|
| $Y$        | 0   | 3   | 8   | 15  | Total |
| $P(Y = y)$ | 0.1 | 0.3 | 0.5 | 0.1 | 1     |

$$\begin{aligned}\therefore P(1 < Y < 10) &= P(y = 3) + P(y = 8) \\ &= 0.3 + 0.5 = 0.8\end{aligned}$$

$$\begin{aligned}\text{Now, mean } \bar{Y} &= \sum p_i y_i \\ &= 0(0.1) + 3(0.3) + 8(0.5) + 15(0.1) \\ &= 0 + 0.9 + 4.0 + 1.5 = 6.4\end{aligned}$$

$$\begin{aligned}E(Y^2) &= \sum p_i y_i^2 \\ &= 0^2(0.1) + 3^2(0.3) + 8^2(0.5) + 15^2(0.1) \\ &= 0 + 2.7 + 32 + 22.5 = 57.2\end{aligned}$$

$$\begin{aligned}\therefore \text{Variance} &= E(Y^2) - (E(Y))^2 \\ &= 57.2 - 40.96 = 16.24\end{aligned}$$



EX.4 A r.v.  $X$  has the following probability distribution

|            |       |       |       |        |        |
|------------|-------|-------|-------|--------|--------|
| $x$        | $-2$  | $-1$  | $0$   | $1$    | $2$    |
| $P(X = x)$ | $1/5$ | $1/5$ | $2/5$ | $2/15$ | $1/15$ |

Find the probability distribution of

(i)  $V = X^2 + 1,$

(ii)  $W = X^2 + 2X + 3$

| $V$    | 1     | 2     | 5      |
|--------|-------|-------|--------|
| $P(V)$ | $2/5$ | $1/3$ | $4/15$ |

| $W$    | 2     | 3     | 6      | 11     |
|--------|-------|-------|--------|--------|
| $P(W)$ | $1/5$ | $3/5$ | $2/15$ | $1/15$ |

EX. If the mean of the following distribution is 16 find  $m$ ,  $n$  and variance

|            |   |               |     |     |               |                |
|------------|---|---------------|-----|-----|---------------|----------------|
| $X$        | : | 8             | 12  | 16  | 20            | 24             |
| $P(X = x)$ | : | $\frac{1}{8}$ | $m$ | $n$ | $\frac{1}{4}$ | $\frac{1}{12}$ |

EX. A function is defined as

$$f(x) = \begin{cases} 0, & \text{for } x < 2 \\ \frac{2x+3}{18}, & \text{for } 2 \leq x \leq 4 \\ 0, & \text{for } x > 4 \end{cases}$$

Show that  $f(x)$  is a probability density function and find the probability that  $2 < x < 3$ .

**EX.** A continuous random variable has probability density function  
 $f(x) = 6(x - x^2), 0 \leq x \leq 1$  Find

- (i) mean      (ii) Variance
- (iii) median      (iv) mode
- (v) harmonic mean
- (vi)  $P(|x - m| < \sigma)$
- (vii)  $P(\mu - 2\sigma < X < \mu + 2\sigma)$

where  $m = \text{median}, \mu = \text{mean}, \sigma = \text{S.D.}$

## Solution:

$$\begin{aligned} \text{(i)} \quad E(X) &= \int_0^1 x \cdot 6(x - x^2) dx \\ &= 6 \int_0^1 (x^2 - x^3) dx \\ &= 6 \left[ \frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 \\ &= 6 \left( \frac{1}{3} - \frac{1}{4} \right) \\ &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad E(X^2) &= \int_0^1 x^2 \cdot 6(x - x^2) dx \\
 &= 6 \int_0^1 (x^3 - 3x^4) dx \\
 &= 6 \left[ \frac{x^4}{4} - \frac{3x^5}{5} \right]_0^1 \\
 &= 6 \left( \frac{1}{4} - \frac{3}{5} \right) \\
 &= \frac{6}{20} = \frac{3}{10}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Var.}(x) &= E(X^2) - [E(X)]^2 \\
 &= \frac{3}{10} - \frac{1}{4} = \frac{1}{20}
 \end{aligned}$$

(iii) Median  $M$  is given by  $\int_0^M 6(x - x^2) dx = \frac{1}{2}$

$$\therefore 6 \left[ \frac{x^2}{2} - \frac{x^3}{3} \right]_0^M = \frac{1}{2}$$

$$\therefore 6 \left[ \frac{M^2}{2} - \frac{M^3}{3} \right] = \frac{1}{2}$$

$$\therefore 3M^2 - 2M^3 = \frac{1}{2}$$

$$\therefore 4M^3 - 6M^2 + 1 = 0$$

$$\therefore (2M - 1)(2M^2 - 2M - 1) = 0$$

$$\therefore M = \frac{1}{2} \text{ or } \frac{2 \pm \sqrt{12}}{4} = \frac{1 \pm \sqrt{3}}{2}$$

Of the three values only  $M = 1/2$  lies in  $(0, 1)$

Hence,  $M = 1/2$



**(iv)** For mode we must have  $f'(x) = 0$

and  $f''(x) < 0$

$$\therefore f'(x) = 6(1 - 2x) = 0$$

$$\therefore x = 1/2$$

$$f''(x) = -12$$

$$\therefore \text{Mode} = \frac{1}{2}$$

**(v)** Harmonic mean is given by

$$\frac{1}{H} = \int_0^1 \frac{1}{x} \cdot 6(x - x^2) dx$$

$$= 6 \int_0^1 (1 - x) dx$$

$$= 6 \left[ x - \frac{x^2}{2} \right]_0^1$$

$$= 6 \left[ 1 - \frac{1}{2} \right] = 3$$

$$\therefore H = \frac{1}{3}$$

$$\begin{aligned}
\text{(vi)} \quad P(|x - M| < \sigma) &= P(-\sigma < x - M < \sigma) \\
&= P(M - \sigma < x < M + \sigma) \\
&= P(0.5 - 0.2236 < x < 0.5 + 0.2230) \\
&= P(0.2764 < x < 0.7236) \\
&= 6 \int_{0.2764}^{0.7236} (x - x^2) dx \\
&= 6 \left[ \frac{x^2}{2} - \frac{x^3}{3} \right]_{0.2764}^{0.7236} \\
&= 6 \left[ \left( \frac{(0.7236)^2}{2} - \frac{(0.7236)^3}{3} \right) \right. \\
&\quad \left. - \left( \frac{(0.2764)^2}{2} - \frac{(0.2764)^3}{3} \right) \right] \\
&= 6[0.1355 - 0.0311] = 0.6264
\end{aligned}$$

EX 5. If  $X$  is a continuous random variable with probability density function given by

$$f(x) = \begin{cases} k(x - x^3); & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find **(i)**  $k$ , **(ii)** mean,  
**(iii)** variance, **(iv)** median.  
**(v)** Mode

EX. A continuous random variable  $X$  has p.d.f.

$$f(x) = kx^2 e^{-x}, x \geq 0.$$

Find  $k$ , mean and variance

 EX. The daily consumption of electric power (in million kwh ) is a random variable  $X$  with probability distribution function

$$f(x) = \begin{cases} kxe^{-x/3}, & \text{for } x > 0 \\ 0, & \text{for } x \leq 0 \end{cases}$$

 Find the value of  $k$ , the expectation of  $X$  and the probability that on a given day the electric consumption is more than expected value.

Ex. Let  $X$  be a continuous random variable with p.d.f.  $f(x) = kx(1 - x)$ ,  $0 \leq x \leq 1$ . Find  $k$  and determine a number  $b$  such that

$$P(X \leq b) = P(X \geq b)$$